CodeAlpha Java Internship
Stock Trading Platform

Java Source Code

```java
import java.util.*;

public class StockTradingPlatform {
    static class Stock {
        String symbol;
        String name;
        double price;

        Stock(String symbol, String name, double price) {
            this.symbol = symbol;
            this.name = name;
            this.price = price;
        }
    }

    static class User {
        String name;
        double cash;
        HashMap<String, Integer> portfolio = new HashMap<>();

        User(String name, double cash) {
            this.name = name;
            this.cash = cash;
        }
    }

    static Scanner sc = new Scanner(System.in);
    static ArrayList<Stock> stocks = new ArrayList<>();
    static User user;

    static Stock findStock(String symbol) {
        for (Stock s : stocks)
            if (s.symbol.equalsIgnoreCase(symbol)) return s;
        return null;
    }

    static void showMarket() {
        System.out.println("\n=== MARKET DATA ===");
        System.out.printf("%-10s %-20s %10s%n", "Symbol", "Company", "Price");
        for (Stock s : stocks)
            System.out.printf("%-10s %-20s %10.2f%n", s.symbol, s.name, s.price);
    }

    static void buy() {
        System.out.print("Enter stock symbol: ");
        String symbol = sc.next();
        Stock s = findStock(symbol);
        if (s == null) {
            System.out.println("Stock not found.");
            return;
        }

        System.out.print("Quantity: ");
        int qty = sc.nextInt();
        if (qty <= 0) {
            System.out.println("Quantity must be positive.");
            return;
        }

        double cost = s.price * qty;
        if (cost > user.cash) {
            System.out.println("Insufficient cash.");
            return;
        }

        user.cash -= cost;
        user.portfolio.put(s.symbol, user.portfolio.getOrDefault(s.symbol, 0) + qty);
        System.out.printf("Bought %d shares of %s for %.2f%n", qty, s.symbol, cost);
    }

    static void sell() {
        System.out.print("Enter stock symbol: ");
        String symbol = sc.next();
        Stock s = findStock(symbol);
        if (s == null) {
            System.out.println("Stock not found.");
            return;
        }

        int owned = user.portfolio.getOrDefault(s.symbol, 0);
        System.out.print("Quantity: ");
        int qty = sc.nextInt();
```

```java
        if (qty <= 0 || qty > owned) {
            System.out.println("Invalid quantity.");
            return;
        }

        double value = s.price * qty;
        user.cash += value;
        int remaining = owned - qty;
        if (remaining == 0) user.portfolio.remove(s.symbol);
        else user.portfolio.put(s.symbol, remaining);

        System.out.printf("Sold %d shares of %s for %.2f%n", qty, s.symbol, value);
    }

    static void showPortfolio() {
        System.out.println("\n=== PORTFOLIO ===");
        double stockValue = 0;

        if (user.portfolio.isEmpty()) {
            System.out.println("No stocks owned.");
        } else {
            for (Map.Entry<String, Integer> e : user.portfolio.entrySet()) {
                Stock s = findStock(e.getKey());
                double value = s.price * e.getValue();
                stockValue += value;
                System.out.printf("%s : %d shares | Value: %.2f%n",
                        s.symbol, e.getValue(), value);
            }
        }

        System.out.printf("Cash: %.2f%n", user.cash);
        System.out.printf("Total Portfolio Value: %.2f%n", user.cash + stockValue);
    }

    public static void main(String[] args) {
        stocks.add(new Stock("AAPL", "Apple", 200));
        stocks.add(new Stock("MSFT", "Microsoft", 420));
        stocks.add(new Stock("GOOG", "Alphabet", 180));
        stocks.add(new Stock("AMZN", "Amazon", 190));

        System.out.print("Enter your name: ");
        String name = sc.nextLine();
        System.out.print("Enter starting cash: ");
        double cash = sc.nextDouble();
        user = new User(name, cash);

        int choice;
        do {
            System.out.println("\n=== STOCK TRADING PLATFORM ===");
            System.out.println("1. View Market");
            System.out.println("2. Buy Stock");
            System.out.println("3. Sell Stock");
            System.out.println("4. View Portfolio");
            System.out.println("5. Exit");
            System.out.print("Choose: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1 -> showMarket();
                case 2 -> buy();
                case 3 -> sell();
                case 4 -> showPortfolio();
                case 5 -> System.out.println("Thank you for using the platform.");
                default -> System.out.println("Invalid choice.");
            }
        } while (choice != 5);

        sc.close();
    }
}
```

Project README

````
# CodeAlpha Stock Trading Platform

A Java console simulation of a simple stock trading environment.

## Features
- Market data display
- Buy and sell operations
- Portfolio tracking
- Cash balance
- Object-oriented classes for stocks and users

## Run
```bash
javac StockTradingPlatform.java
java StockTradingPlatform
```

## Note
This is an educational simulation. Prices are fixed sample values and no real trading occurs.
````